

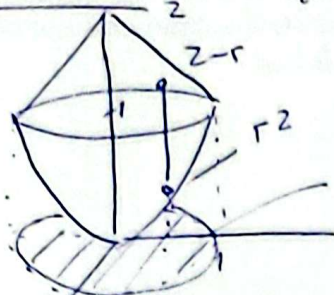
1) $M = \{ \bar{x} \in \mathbb{R}^3 / x^2 + y^2 \leq z \leq 2 - \sqrt{x^2 + y^2} \}$

$z = x^2 + y^2$ paraboloide, $z = 2 - \sqrt{x^2 + y^2}$ semicono inf

proycción semicono

$z = 2 - z \rightarrow 2z = 2 \rightarrow z = 1$

arco de $r=1$ en $z=1$.



$D_{xy} = \begin{cases} x = r \cos \theta & 0 \leq \theta \leq 2\pi \\ y = r \sin \theta & 0 \leq r \leq 1 \\ z = z & r^2 \leq z \leq 2-r \end{cases}$

a) $\text{vol}(M) = \int_0^{2\pi} \int_0^1 \int_{r^2}^{2-r} r \, dz \, dr \, d\theta = 2\pi \int_0^1 r [2-r-r^2] \, dr = 2\pi \int_0^1 (2r - r^2 - r^3) \, dr = 2\pi \left[r^2 - \frac{r^3}{3} - \frac{r^4}{4} \right]_0^1 = 2\pi \left[1 - \frac{1}{3} - \frac{1}{4} \right] = 2\pi \cdot \frac{5}{12} = \frac{5}{6}\pi$

b) $a(D_{xy}) = \int_0^{2\pi} \int_0^1 r \, dr \, d\theta = \int_0^{2\pi} \left[\frac{r^2}{2} \right]_0^1 d\theta = \int_0^{2\pi} \frac{1}{2} d\theta = 2\pi \cdot \frac{1}{2} = \pi$

2) $\bar{g}: \mathbb{R} \rightarrow \mathbb{R}^2 \quad \bar{g}(t) = (e^{t-1} + 2t, t^2 - 2) \quad F: \mathbb{R}^2 \rightarrow \mathbb{R} \quad F(x, y) = y \sin x + e^x y + y$

a) $\text{calcular } JH(0,0) = J\bar{g}(F(0,0)) = J\bar{g}(1) \cdot JF(0,0) = \begin{bmatrix} e^{t-1} + 2 \\ 2t \end{bmatrix}_{t=1} \cdot \begin{bmatrix} y \cos x + y e^x + 1 \\ \sin x + x e^x + 1 \end{bmatrix}_{(0,0)} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 3 \\ 0 & 2 \end{bmatrix}$

b) $\nabla H(0,0) = (0, 3) \quad \|\nabla H(0,0)\| = (9, 1) \quad \vec{v}_{\text{uni}_1} = (1, 0) ; \vec{v}_{\text{uni}_2} = (-1, 0) \quad (0,0)$

3) $F = (xy, \frac{x^2}{2} + 3x - xy) \quad P_x = x + 3 - y \neq P_y = x \quad F$ no conservativo.

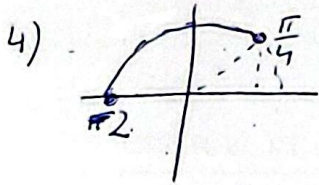
$\sigma: [\pi, 2\pi] \rightarrow \mathbb{R}^2 / \sigma(t) = (2 \cos t, 2 \sin t) \quad \sigma'(t) = (-2 \sin t, 2 \cos t)$
 $W(F) = \int_{\pi}^{2\pi} (4 \cos t \sin t, 2 \cos^2 t + 3 \sin t - 4 \cos t \sin t) \cdot (-2 \sin t, 2 \cos t) \, dt$
 $= \int_{\pi}^{2\pi} (-8 \cos t \sin^2 t + 4 \cos^3 t + 6 \sin t \cos t - 8 \cos^2 t \sin t) \, dt$
 Salen por sustitución y sabiendo que $\int_{\pi}^{2\pi} \cos^3 t \, dt = 0$.

Teorema de Green $\oint_R (3 - y) \, dA = \int_0^2 \int_{\pi}^{2\pi} (3 - 2 \sin t) \, dt \, dr = \int_0^2 \left[3t + 2 \cos t \right]_{\pi}^{2\pi} \, dr = \int_0^2 (6\pi - 4) \, dr = [6\pi r - 4r]_0^2 = 12\pi - 8$

segmento $\varphi: [-2, 2] \rightarrow \mathbb{R}^2 / \varphi(t) = (t, 0) \quad \varphi'(t) = (1, 0)$
 $\int_{\varphi} F(\varphi) \varphi'(t) \, dt = \int_{-2}^2 (0, \frac{t^2}{2} + 3t) \cdot (1, 0) \, dt = \int_{-2}^2 0 \, dt = 0$

$\therefore \int_C \vec{F} \, d\vec{s} = 12\pi - 8$

b) Falsa pues el campo no es conservativo.



a) $C = \text{Im}(\sigma)$, $\sigma: [\pi/4, \pi] \rightarrow \mathbb{R}^2$

$\sigma(t) = (2 \cos t, 2 \sin t)$

$\|\sigma'(t)\| = 2$

$$L(C) = \int_{\pi/4}^{\pi} 2 \, dt = 2t \Big|_{\pi/4}^{\pi} = 2 \left(\pi - \frac{\pi}{4} \right) = 2 \cdot \frac{3\pi}{4} = \frac{3}{2} \pi$$

b) $M(C) = \int_{\pi/4}^{\pi} F(\sigma(t)) \|\sigma'(t)\| \, dt = \int_{\pi/4}^{\pi} 2 \sin t \cdot 4 \cos^2 t \, dt =$

$$= 16 \cdot \int_{\pi/4}^{\pi} \cos^2 t \sin t \, dt = -16 \int_{\frac{\sqrt{2}}{2}}^{-1} u^2 \, du = -16 \frac{u^3}{3} \Big|_{\frac{\sqrt{2}}{2}}^{-1} = 16 \left[\frac{\sqrt{2}}{3} + \frac{1}{3} \right]$$

⑤ $\begin{cases} u = \cos t \\ du = -\sin t \, dt \end{cases}$